



Proof · Mark Scheme

Chapter 1: Proof. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Proof by deduction

- 1** Let the integers be $n, n+1, n+2$ where n is an integer *B1*. Sum = $3n+3 = 3(n+1)$ *M1*. $n+1$ is an integer, so the sum is a multiple of 3 *A1*.
- 2** Let the odd numbers be $2m+1$ and $2n+1$, m, n integers (different letters) *B1*. $(2m+1)(2n+1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1$ *M1*. $2mn + m + n$ is an integer, so the product is odd *A1*.
- 3** $x^2 - 6x + 10 = (x-3)^2 + 1$ *M1 A1*. $(x-3)^2 \geq 0$ for all real x , so $(x-3)^2 + 1 \geq 1 > 0$ *A1*.
- 4** Consecutive odd numbers $2n+1$ and $2n+3$ *B1*. $(2n+3)^2 - (2n+1)^2 = (4n^2 + 12n + 9) - (4n^2 + 4n + 1)$ *M1* = $8n + 8$ *A1* = $8(n+1)$, a multiple of 8 since $n+1$ is an integer *A1*.
- 5(a)** Start from a true statement: $(\sqrt{x} - \sqrt{y})^2 \geq 0$ (valid since $x, y > 0$ so roots are real) *M1*. Expand: $x - 2\sqrt{xy} + y \geq 0$ *M1* $\Rightarrow x + y \geq 2\sqrt{xy}$ *A1* $\Rightarrow \sqrt{xy} \leq \frac{x+y}{2}$ for all positive x, y *A1 – concluding statement*.
Note: starting from the result and working to a true statement scores at most M1 M1 unless the argument is clearly reversible.
- 5(b)** E.g. $x = -1, y = -4$: $\sqrt{xy} = \sqrt{4} = 2$ and $\frac{x+y}{2} = -2.5$ *M1*. $2 > -2.5$, so the statement is false *A1*.

Section B – Proof by exhaustion

- 6** $n = 1, \dots, 6$ give 13, 17, 23, 31, 41, 53 *M1*; all are prime, so true for $1 \leq n \leq 6$ *A1*.
- 7** Primes between 20 and 40: 23, 29, 31, 37 *B1*. $23 = 6(4) - 1$, $29 = 6(5) - 1$, $31 = 6(5) + 1$, $37 = 6(6) + 1$ *M1*. All primes checked, so result proved *A1*.
- 8** The last digit of n^2 depends only on the last digit of n *B1*. Last digits 0, 1, ..., 9 square to numbers ending 0, 1, 4, 9, 6, 5, 6, 9, 4, 1 *M1*. None ends in 2, 3, 7 or 8, and all cases are covered *A1*.
- 9** Every integer is $3m, 3m+1$ or $3m+2$ *B1*. $(3m)^2 = 3(3m^2)$; $(3m+1)^2 = 3(3m^2 + 2m) + 1$ *M1 A1*; $(3m+2)^2 = 9m^2 + 12m + 4 = 3(3m^2 + 4m + 1) + 1$. All cases give $3k$ or $3k+1$ *A1*.
- 10(a)** n not a multiple of 3, so $n = 3m+1$ or $n = 3m+2$ *B1*. $(3m+1)^2 - 1 = 9m^2 + 6m = 3(3m^2 + 2m)$; $(3m+2)^2 - 1 = 9m^2 + 12m + 3 = 3(3m^2 + 4m + 1)$ *M1*. Both multiples of 3, so $n^2 - 1$ is a multiple of 3 *A1*.
- 10(b)** $p > 3$ prime, so p is not a multiple of 3; by (a), $3 \mid p^2 - 1$ *B1*. p is odd, so $p^2 - 1 = (p-1)(p+1)$ is a product of two consecutive even numbers *M1*; one of them is a multiple of 4, so the product is a multiple of $2 \times 4 = 8$ *A1*. 3 and 8 have no common factor, so $p^2 - 1$ is a multiple of 24 *A1*.

Section C – Disproof by counterexample

- 11** E.g. $2+3=5$ *M1*, which is odd, so the statement is false *A1*.
- 12** E.g. $x = -1$: $(x+1)^2 = 0$ and $x^2 = 1$ *M1*; $0 < 1$, so the statement is false *A1*. (Any $x \leq -\frac{1}{2}$ works.)
- 13** $n = 1, 2, 3, 4$ give 5, 11, 29, 83 (all prime); $n = 5$: $3^5 + 2 = 245$ *M1* = 5×49 , not prime, so Sam is wrong *A1*.
- 14** E.g. $a = 1, b = -2$: $a > b$ but $a^2 = 1 < 4 = b^2$ *M1 A1*.
- 15(a)** Choose $n = 41$: $41^2 - 41 + 41 = 41^2$ *M1*, which has factor 41, so is not prime *A1*.
- 15(b)** E.g. $x = y = \sqrt{2}$ (irrational) *M1*; $xy = 2$, which is rational, so the statement is not always true *A1*.

Section D – Proof by contradiction

- 16** Assume, for contradiction, that there is a largest even integer N *B1*. $N+2$ is even and $N+2 > N$ *M1*. This contradicts N being the largest, so no largest even integer exists *A1*.



17 Assume, for contradiction, n^2 is odd and n is even *B1*. $n = 2k \Rightarrow n^2 = 4k^2 = 2(2k^2)$, which is even *M1*. Contradicts n^2 odd, so n must be odd *A1*.

18 Assume, for contradiction, that integers a, b exist with $6a + 9b = 1$ *B1*. $3(2a + 3b) = 1 \Rightarrow 2a + 3b = \frac{1}{3}$ *M1*. But $2a + 3b$ is an integer and $\frac{1}{3}$ is not: contradiction, so no such integers exist *A1*.

19 Assume, for contradiction, that positive integers x, y exist with $x^2 - y^2 = 1$ *B1*. $(x - y)(x + y) = 1$ *M1*. Both factors are integers and $x + y > 0$, so $x - y = x + y = 1$ *M1*. Then $y = 0$, contradicting y positive; so no such x, y exist *A1*.

20 Assume, for contradiction, that there are finitely many primes $p_1, p_2, \dots, p_n$ *B1*. Let $N = p_1 p_2 \cdots p_n + 1$ *M1*. N leaves remainder 1 when divided by each p_i , so none of $p_1, \dots, p_n$ divides N *M1*. So N is either prime itself or has a prime factor not in the list; either way there is a prime not in the list *A1*. This contradicts the list containing all primes, so there are infinitely many primes *A1*.

Common error: claiming N itself must be prime. (E.g. $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1 = 30031 = 59 \times 509$.)

Roar-level challenge

21 $n^3 - n = (n - 1)n(n + 1)$ *M1*, a product of three consecutive integers *A1*. One of any two consecutive integers is even, so the product is a multiple of 2; one of any three consecutive integers is a multiple of 3, so the product is a multiple of 3 *M1*. 2 and 3 have no common factor, so $n^3 - n$ is a multiple of 6 *A1*.

Alternative: exhaustion on $n \bmod 2$ and $n \bmod 3$.